

Client: Sydney Water Corporation
Project: Willoughby Submain Amplification Project
Location: Matthew Street, Crows Nest, NSW 2065

Project No: 60764076
Logged by: KM
Start Date: 06/07/26
Checked by: RH
End Date: 09/07/26
Location Meth.: RTK
RL: 80.29 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 51.23 m
Hole Diameter: 96-300 mm
Inclination: 90°
Bearing: N/A

Easting: 333834.10
Northing: 6256025.30
Hor. Datum: MGA2020-56

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
D T					0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 100 mm thick; road surface.			PAVEMENT
VE	U	ES: 0.15-0.25 m PID=1.3 PPM D: 0.25-0.35 m ES: 0.30-0.40 m PID=0.9 PPM			84.0			SANDSTONE: medium to coarse grained, pale yellow brown mottled red brown. from 0.40-0.50 m: ironstone band, red brown.			HAWKESBURY SANDSTONE Vacuum excavation through highly weathered rock to confirm absence of boulders
					1.0						
					83.0						
					2.0						
					82.0						
					3.0						
					81.0						
					4.0						
					80.0						
					5.0						
					79.0						
					6.0						
					78.0						
					7.0						
					77.0						
					8.0						

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METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (ROD) (%)	INFERRED STRENGTH										DEFECT SPACING (mm)				ADDITIONAL OBSERVATIONS (Defect Descriptions)																																								
										20 x IS ₍₅₀₎ A: ● D: ○ UCS: □																																																						
										SOIL	VL	L	M	H	VS	VH	EH	20	60	200	600	2000																																										
HQ3	1	3.70	Is(50) D=1 MPa Is(50) A=1.27 MPa	83.0	1.0		SANDSTONE: medium to coarse grained, pale grey, indistinctly thinly bedded. NO CORE: 1.35-2.05 m.	MW to SW	73 (73) [67]									HAWKESBURY SANDSTONE 1.14-1.28 m: DB, broken core from reaming in casing, 170 mm thick																																														
																			Is(50) D=1.01 MPa Is(50) A=1.14 MPa	82.0	SANDSTONE: medium to coarse grained, red brown and brown grey, trace black specks, trace fine, angular quartz gravels, thickly bedded with distinct thin laminations at 5-10°; iron staining.	MW to SW									2.17 m: P, 15°, RF, IR, SN Fe 2.25 m: P, 15°, RF, IR, CN, 15 mm aperture 2.53-2.60 m: P, 5-10°, RF, IR, CN, x 2 2.70 m: HB 2.76 m: P, 10°, RF, CU, CN 2.91 m: P, 20°, RF, CU, SN Fe 3.00 m: HB 3.31 m: P, 10°, RF, VN CL 3.41-3.43 m: P, 5°, RF, PR, CT Cl, x 2 3.57 m: DB 3.79 m: DB 3.90-3.96 m: P, 5-10°, RF, PR, CN, x 2 4.00 m: HB 4.10-4.18 m: DB, x 5 4.30-4.34 m: P, 0-15°, RF, PR, CN, x 2 4.92 m: DB 5.00 m: HB 5.22-5.40 m: DB, x 3 5.31-5.38 m: DB, x 2 5.71-5.81 m: P, 10°, RF, PR, CN, x 3 5.87-5.93 m: DB, x 3 6.00 m: HB 6.28 m: J, 90°, VR, PR, CN, 150 mm persistence 6.28-6.33 m: EW, 0°, RF, PR, SP, 50 mm thick 6.71 m: DB 6.78 m: DB 6.82 m: P, 5°, RF, IR, CN 6.95 m: P, 5°, RF, PR, VN CL 7.00 m: HB 7.15-7.19 m: EW, 5-15°, RF, PR, SP, 40 mm thick 7.36 m: DB 7.59 m: P, 15°, RF, PR, CN 7.71-7.75 m: P, 10-20°, RF, IR, CN, x 2 7.94 m: P, 5° RF, PR, CN																																	
																																Is(50) D=0.43 MPa Is(50) A=0.39 MPa	81.0	SANDSTONE: medium to coarse grained, red brown and brown grey, trace black specks, trace fine, angular quartz gravels, thickly bedded with distinct thin laminations at 5-10°; iron staining.	MW to SW									2.17 m: P, 15°, RF, IR, SN Fe 2.25 m: P, 15°, RF, IR, CN, 15 mm aperture 2.53-2.60 m: P, 5-10°, RF, IR, CN, x 2 2.70 m: HB 2.76 m: P, 10°, RF, CU, CN 2.91 m: P, 20°, RF, CU, SN Fe 3.00 m: HB 3.31 m: P, 10°, RF, VN CL 3.41-3.43 m: P, 5°, RF, PR, CT Cl, x 2 3.57 m: DB 3.79 m: DB 3.90-3.96 m: P, 5-10°, RF, PR, CN, x 2 4.00 m: HB 4.10-4.18 m: DB, x 5 4.30-4.34 m: P, 0-15°, RF, PR, CN, x 2 4.92 m: DB 5.00 m: HB 5.22-5.40 m: DB, x 3 5.31-5.38 m: DB, x 2 5.71-5.81 m: P, 10°, RF, PR, CN, x 3 5.87-5.93 m: DB, x 3 6.00 m: HB 6.28 m: J, 90°, VR, PR, CN, 150 mm persistence 6.28-6.33 m: EW, 0°, RF, PR, SP, 50 mm thick 6.71 m: DB 6.78 m: DB 6.82 m: P, 5°, RF, IR, CN 6.95 m: P, 5°, RF, PR, VN CL 7.00 m: HB 7.15-7.19 m: EW, 5-15°, RF, PR, SP, 40 mm thick 7.36 m: DB 7.59 m: P, 15°, RF, PR, CN 7.71-7.75 m: P, 10-20°, RF, IR, CN, x 2 7.94 m: P, 5° RF, PR, CN																				
																																													Is(50) D=1.46 MPa Is(50) A=1.41 MPa	80.0	SANDSTONE: medium to coarse grained, red brown and brown grey, trace black specks, trace fine, angular quartz gravels, thickly bedded with distinct thin laminations at 5-10°; iron staining.	MW to SW									2.17 m: P, 15°, RF, IR, SN Fe 2.25 m: P, 15°, RF, IR, CN, 15 mm aperture 2.53-2.60 m: P, 5-10°, RF, IR, CN, x 2 2.70 m: HB 2.76 m: P, 10°, RF, CU, CN 2.91 m: P, 20°, RF, CU, SN Fe 3.00 m: HB 3.31 m: P, 10°, RF, VN CL 3.41-3.43 m: P, 5°, RF, PR, CT Cl, x 2 3.57 m: DB 3.79 m: DB 3.90-3.96 m: P, 5-10°, RF, PR, CN, x 2 4.00 m: HB 4.10-4.18 m: DB, x 5 4.30-4.34 m: P, 0-15°, RF, PR, CN, x 2 4.92 m: DB 5.00 m: HB 5.22-5.40 m: DB, x 3 5.31-5.38 m: DB, x 2 5.71-5.81 m: P, 10°, RF, PR, CN, x 3 5.87-5.93 m: DB, x 3 6.00 m: HB 6.28 m: J, 90°, VR, PR, CN, 150 mm persistence 6.28-6.33 m: EW, 0°, RF, PR, SP, 50 mm thick 6.71 m: DB 6.78 m: DB 6.82 m: P, 5°, RF, IR, CN 6.95 m: P, 5°, RF, PR, VN CL 7.00 m: HB 7.15-7.19 m: EW, 5-15°, RF, PR, SP, 40 mm thick 7.36 m: DB 7.59 m: P, 15°, RF, PR, CN 7.71-7.75 m: P, 10-20°, RF, IR, CN, x 2 7.94 m: P, 5° RF, PR, CN							
																																																										Is(50) D=0.48 MPa Is(50) A=0.68 MPa	79.0	SANDSTONE: medium to coarse grained, red brown and brown grey, trace black specks, trace fine, angular quartz gravels, thickly bedded with distinct thin laminations at 5-10°; iron staining.	MW to SW			

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METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L 6 M 20 H 60 VH 200 EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	8.70	5	Is(50) D=1.08 MPa Is(50) A=0.94 MPa	76.0			SANDSTONE: medium grained, pale grey, speckled black, very thinly bedded at 5-15° with grey laminations, trace black specks.	MW to SW				8.00 m: HB 8.21 m: DB 8.37 m: P, 0°, RF, PR, SN Fe
	9.33	6	Is(50) D=1.27 MPa Is(50) A=1.25 MPa C: 10.15-10.40 m UCS=26 MPa Is(50) A=1.27 MPa Is(50) D=1.12 MPa	9.0			SANDSTONE: medium to coarse grained, red brown, with black specks, indistinctly thinly bedded with laminations at 10-20°; iron staining.	SW	100 (100) [100]			8.65 m: P, 5°, RF, PR, CN 8.67 m: DB 8.86-8.92 m: DB, x 2 9.00 m: HB 9.06-9.43 m: DB, x 7
	12.33	7	Is(50) D=0.45 MPa Is(50) A=0.5 MPa	75.0			from 9.33 m: becoming yellow brown and light brown, very thinly cross bedded with laminations.		100 (91) [80]			9.52 m: DB 9.89 m: DB 10.00 m: HB
	13.84	8	Is(50) A=1.24 MPa Is(50) D=1.15 MPa	74.0					68 (68) [52]			10.48-10.81 m: P, 0-10°, RF, PR, CN, x 3 10.85 m: J, 80°, RF, IR, SN Fe, 270 mm persistence 11.00 m: HB 11.23-11.46 m: DB, fractured rock along sub-vertical joint, partially healed 11.28 m: J, 80°, RF, PR, SN Fe, 340 mm persistence 11.64 m: P, 5°, RF, PR, CN 11.68 m: DB 11.89 m: DB 12.00 m: HB 12.10-12.57 m: DB, x 3 12.22 m: P, 5-10°, RF, UN, CN
	15.33		Is(50) A=1.43 MPa Is(50) D=1.17 MPa	73.0								12.68-12.90 m: P, 5-10°, RF, PR, CN, x 4 12.85 m: J, 85°, RF, CU, SN Fe, 140 mm persistence 13.00 m: HB 13.06-13.16 m: P, 0-5°, RF, PR, CN, x 2 13.25 m: J, 70°, RF, PR, SN Fe, 200 mm persistence
			Is(50) A=0.29 MPa Is(50) D=0.32 MPa	72.0			NO CORE: 13.35-13.84 m.					13.92 m: DB 14.00 m: HB 14.16-14.30 m: DB 14.33-14.82 m: DB, x 7
			Is(50) A=0.98 MPa Is(50) D=0.86 MPa	71.0			SANDSTONE: medium grained, pale grey and red, very thickly bedded with distinct laminations at 5-30°, with black specks.	MW to SW	100 (100) [100]			14.98-15.00 m: HB 15.24 m: DB 15.41-15.48 m: DB, x 2
				70.0				SW to FR				
				69.0					100 (100) [100]			
				68.0								

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HQ3	9		Is(50) D=0.83 MPa Is(50) A=1.19 MPa C: 16.25-16.40 m C: 16.40-16.60 m UCS=18 MPa	68.0			SANDSTONE: medium grained, pale grey and red, very thickly bedded with distinct laminations at 5-30°, with black specks.	SW to FR				16.00 m: HB 16.23 m: P, 10°, RF, PR, CN
			Is(50) A=1.08 MPa Is(50) D=2.03 MPa	67.0			from 17.44 m: becoming red brown. from 17.45-17.90 m: ironstone bands and iron staining.					16.73 m: P, 15°, RF, PR, CN 16.80 m: DB 17.00 m: HB 17.43 m: DB
	10		Is(50) A=1.48 MPa Is(50) D=1.57 MPa	66.0			from 18.90 m: trace fine, sub-rounded quartz gravels.		100 (100) [100]			18.00 m: HB 18.62 m: HB 18.97-19.00 m: HB
HQ3	11		Is(50) A=1.54 MPa Is(50) D=1.29 MPa	65.0								20.00 m: HB 20.07 m: P, 20°, RF, UN, SN Fe
			Is(50) A=0.97 MPa Is(50) D=0.19 MPa	64.0								20.81 m: HB 20.97-21.00 m: HB 21.00 m: HB 21.08 m: HB
			Is(50) A=0.23 MPa Is(50) D=0.21 MPa	63.0			SHALE: purple and grey, rip up clast. SANDSTONE: medium to coarse grained, red brown, distinctly laminated at 5 to 20°. from 21.20 m: becoming distinctly laminated at 20°. from 21.68 m: becoming pale grey.	HW MW	100 (96) [84]			21.94 m: DB 21.98 m: HB 22.00 m: HB 22.34 m: DB 22.52-22.54 m: EW, 0°, SM, PR, CH, 20 mm thick 22.68 m: P, 5°, RF, IR, VN CL
HQ3	11		PP=100, 150, 160 kPa PP=80, 80 kPa Is(50) A=0.29 MPa Is(50) D=0.24 MPa C: 22.80-23.00 m UCS=1 MPa C: 23.00-23.15 m	62.0			LAMINITE: shale (70%), dark grey, interlaminated at 0-5° with sandstone (30%), fine grained, grey, thinly laminated.	MW to SW				23.00 m: HB 23.23-23.33 m: DB, x 2 23.36-23.40 m: EW, 0-5°, SM, PR, CI, 40 mm thick 23.49-23.52 m: P, 5°, SM, PR, CN, x 2 23.61 m: J, 80°, RF, PR, CN, 400 mm persistence
				61.0								23.92-24.00 m: HB
				24.0								

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HQ3	24.30		Is(50) D=0.12 MPa Is(50) A=0.22 MPa Is(50) D=0.48 MPa Is(50) A=0.43 MPa	60.0			LAMINITE: shale (70%), dark grey, interlaminated at 0-5° with sandstone (30%), fine grained, grey, thinly laminated. SANDSTONE: medium to coarse grained, pale grey, very thinly bedded at 0-25°, with grey laminations.	MW to SW FR	100 (100) [102]			24.04 m: P, 5°, SM, PR, VN CI 24.13 m: P, 5°, RF, PR, CN
			Is(50) A=0.86 MPa Is(50) D=0.99 MPa	59.0								24.84 m: DB 25.00 m: HB 25.03-25.04 m: HB, x 2 25.32 m: HB
			Is(50) A=1.05 MPa Is(50) D=1.28 MPa C: 26.30-26.50 m UCS=22 MPa C: 26.50-26.65 m	58.0								25.86 m: P, 5°, RF, UN, CN 26.00 m: HB 26.66 m: DB
	27.20		Is(50) A=0.61 MPa Is(50) D=0.42 MPa	57.0			from 27.20 m: pale pink grey, very thinly cross bedded, with grey laminations	SW to FR	100 (96) [95]			27.00 m: HB 27.06 m: P, 10°, RF, PR, CN 27.15 m: P, 5°, RF, PR, SN Fe
				56.0								28.00 m: DB
			Is(50) A=0.87 MPa Is(50) D=0.93 MPa	55.0								28.69 m: P, 30°, RF, PR, CN 28.88 m: DB 29.00-29.07 m: P, 5°, RF, PR, SN Fe, x 3
			Is(50) D=0.62 MPa Is(50) A=1.04 MPa	30.0				MW				29.43 m: P, 20°, RF, PR, VN CL 29.69 m: J, 30°, RF, CU, SN Fe, 10 mm persistence 29.77-29.80 m: EW, 0-5°, RF, PR, SC, 30 mm thick 29.82-29.84 m: EW, 0-5°, RF, PR, SP, 20 mm thick 29.89-31.24 m: P, 5-20°, RF, PR to UN, CN to SN Fe, x 6 29.90-29.96 m: SS, 10-40°, 60 mm thick 30.02 m: J, 70°, RF, CU, SN Fe, 170 mm persistence 30.05-30.16 m: DB, x 2 30.43 m: J, 70-85°, RF, CU, SN Fe, 300 mm persistence 30.63 m: DB 30.66 m: J, 80°, RF, IR, SN Fe, 60 mm persistence, healed 30.74 m: J, 70°, VR, CU, VN Fe, 140 mm thick 31.00 m: HB 31.05 m: J, 70°, RF, PR, CN, 100 mm persistence 31.10 m: DB 31.33 m: J, 20°, SM, IR, CT CI, 60 mm persistence 31.52 m: J, 70°, RF, CU, VN CL, 90 mm persistence
	30.20		Is(50) D=0.94 MPa Is(50) A=1.51 MPa	53.0			from 31.20-31.50 m: becoming very thinly cross bedded with very thin laminations.	XW	98 (97) [70]			31.71 m: DB 31.79-31.86 m: EW, 10-15°, RF, GP, 70 mm thick
			Is(50) A=0.76 MPa Is(50)	32.0			SANDSTONE: medium to coarse grained, pale grey, with dark brown flecks. NO CORE: 31.86-31.95 m. SANDSTONE: medium to coarse grained, pale grey, massive, with dark brown flecks.	MW				

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HQ3	33.20	15	D=0.74 MPa Is(50) A=0.74 MPa Is(50) D=0.59 MPa	52.0			SANDSTONE: medium to coarse grained, pale grey, massive, with dark brown flecks.	SW to FR	100 (98) [91]			32.00 m: HB 32.02-32.27 m: P, 5-10°, RF, PR, CN, x 3		
													32.84-33.00 m: HB, x 2	
			Is(50) A=0.7 MPa Is(50) D=0.92 MPa	33.0									33.57-33.62 m: P, 0-5°, RF, PR, CT Cl, x 2 33.66 m: DB 33.69 m: P, 10°, RF, PR, CN 33.77 m: J, 20°, RF, CU, CN, 20 mm persistence 33.83 m: DB 34.00-34.05 m: HB 34.11 m: DB	
				Is(50) A=1.6 MPa Is(50) D=1.88 MPa C: 34.38-34.60 m UCS=28 MPa C: 34.60-34.75 m	50.0		SANDSTONE: medium to coarse grained, red brown, trace ironstone bands, indistinctly bedded with thin laminations; iron staining.					35.00 m: HB 35.03 m: P, 15°, RF, UN, CN 35.14-35.32 m: P, 5-10°, SM, PR, VN Cl, x 3		
				Is(50) A=0.8 MPa Is(50) D=0.97 MPa	49.0		SANDSTONE: medium to coarse grained, pale grey, trace ironstone bands, distinctly very thinly bedded with laminations at 20°.					35.49 m: J, 55°, RF, CU, CN, 60 mm persistence 35.81-35.88 m: EW, 10°, RF, CU, CL, 70 mm thick 35.98-36.00 m: HB		
					36.0									
36.20	16		Is(50) D=0.83 MPa Is(50) A=1.31 MPa	48.0			SANDSTONE: medium to coarse grained, red brown and brown, with fine to medium, sub-rounded quartz gravels, very thinly cross bedded with laminations; iron staining.	SW	100 (99) [70]			36.20 m: J, 55°, VR, CU, SN Fe, 110 mm persistence 36.25 m: P, 10°, RF, CU, CN 36.44-36.93 m: DB, x 3 36.66 m: J, 65°, RF, PR, SN Fe, 150 mm persistence		
													37.00 m: HB 37.03 m: J, 50°, RF, ST, SN Fe, 60 mm persistence 37.07 m: DB 37.14 m: J, 80°, CU, SN Fe, healed, 120 mm persistence 37.17-37.43 m: DB, x 2 37.62-37.63 m: EW, 0-5°, RF, PR, CL, 10 mm thick 37.72 m: DB	
														37.90 m: J, 80°, RF, CU, SN Fe, 90 mm persistence 38.00 m: HB 38.01 m: J, 80°, RF, ST, SN Fe, 450 mm persistence 38.25 m: J, 30°, RF, PR, CN, 20 mm persistence 38.35 m: P, 5°, RF, PR, SN Fe 38.45 m: P, 0°, RF, PR, CN
														38.66 m: J, 70°, RF, CU, SN Fe
39.21			Is(50) D=1.57 MPa Is(50) A=1.52 MPa	39.0			SANDSTONE: medium to coarse grained, red brown and grey, trace fine, sub-rounded, quartz gravels., very thinly cross bedded with indistinct laminations.		100 (99) [98]			38.89 m: DB 39.00 m: HB 39.05-39.14 m: DB, x 2		
			Is(50) D=1.74 MPa Is(50) A=1.82 MPa C: 39.65-39.85 m	45.0										

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Location Meth.: RTK
Ver. Datum: AHD
Surface: Pavement (Asphalt)

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 51.23 m
Hole Diameter: 96-300 mm
Inclination: 90°
Bearing: N/A

Easting: 333834.10
Northing: 6256025.30
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • IS ₍₅₀₎ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	17		UCS=28 MPa C: 39.85-40.00 m Is(50) A=1.11 MPa Is(50) D=1.27 MPa	44.0			SANDSTONE: medium to coarse grained, red brown and grey, trace fine, sub-rounded, quartz gravels., very thinly cross bedded with indistinct laminations.	SW				40.00 m: HB
							SANDSTONE: medium to coarse grained, pale grey, very thinly cross bedded with carbonaceous laminations.	FR				40.26-40.30 m: P, 5°, RF, PR, SN Fe, x 2 40.37-40.39 m: EW, 10-20°, RF, PR, Cl, 20 mm thick 40.43 m: DB 40.59 m: P, 0°, RF, PR, SN X 40.73 m: P, 20°, RF, PR, CN
				41.0								41.00 m: HB
				43.0								
	18		Is(50) D=1.12 MPa Is(50) A=1.1 MPa Is(50) A=0.83 MPa Is(50) D=0.99 MPa	42.0					100 (100) [100]			41.57 m: HB 42.00 m: HB 42.37-42.38 m: EW, 0°, RF, PR, SP, 10 mm thick 42.76 m: DB 43.00 m: HB
			Is(50) A=1.09 MPa Is(50) D=1.23 MPa	43.0								43.36 m: P, 10°, SM, PR, CT Cl 43.42 m: DB
				41.0								
			C: 44.00-44.15 m C: 44.15-44.35 m UCS=17 MPa	44.0			from 44.00 m: becoming thinly bedded.					44.00 m: HB 44.57 m: HB 44.77 m: DB 45.00 m: HB
	18		Is(50) D=0.7 MPa Is(50) A=0.77 MPa	45.0					100 (100) [91]			45.54 m: P, 0°, RF, PR, CN 45.88 m: P, 10°, RF, PR, CN 46.00 m: HB 46.27 m: DB
				39.0								
			Is(50) D=0.84 MPa Is(50) A=0.95 MPa Is(50) D=0.84 MPa C: 46.05-47.00 m	46.0								47.00 m: HB 47.10 m: HB
			Is(50) A=1.06 MPa UCS=15 MPa	47.0								47.34 m: P, 5°, RF, PR, CN
			Is(50) A=0.36 MPa Is(50) D=0.34 MPa	47.0								
				48.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water Corporation
Project: Willoughby Submain Amplification Project
Location: Matthew Street, Crows Nest, NSW 2065

Project No: 60764076
Logged by: KM
Start Date: 06/07/26

Checked by: RH
End Date: 09/07/26
Location Meth.: RTK
RL: 80.29 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

Driller: Terratest	Hole Diameter: 96-300 mm
Drill Rig: Comacchio GEO 305	Inclination: 90°
Total Depth: 51.23 m	Bearing: N/A

Easting: 333834.10
Northing: 6256025.30
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x 15 ₍₅₀₎ A: ● D: ○ UCS: □ σ ₂₀	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	48.19		PP=400 kPa Is(50) D=0.4 MPa Is(50) A=1.26 MPa Is(50) A=0.96 MPa Is(50) D=1.13 MPa Is(50) A=0.89 MPa Is(50) D=0.77 MPa PP=150, 160 kPa PP=170, 170, 200 kPa PP=420, 400 kPa Is(50) A=0.25 MPa Is(50) D=0.54 MPa				SANDSTONE: medium to coarse grained, pale grey, very thinly cross bedded with carbonaceous laminations.	FR	100 (100) [98]			47.96 m: J, 60°, RF, CU, CN, 280 mm persistence 48.00 m: HB
	19											
							CL Sandy CLAY: low plasticity, grey; sand, fine to medium grained; with fine to coarse XW gravel inclusions (some rock structure intact). SANDSTONE: medium to coarse grained, pale grey, very thinly cross bedded with carbonaceous laminations.	XW HW				
							Terminated at 51.23 m. Vibrating wire piezometer installed.					

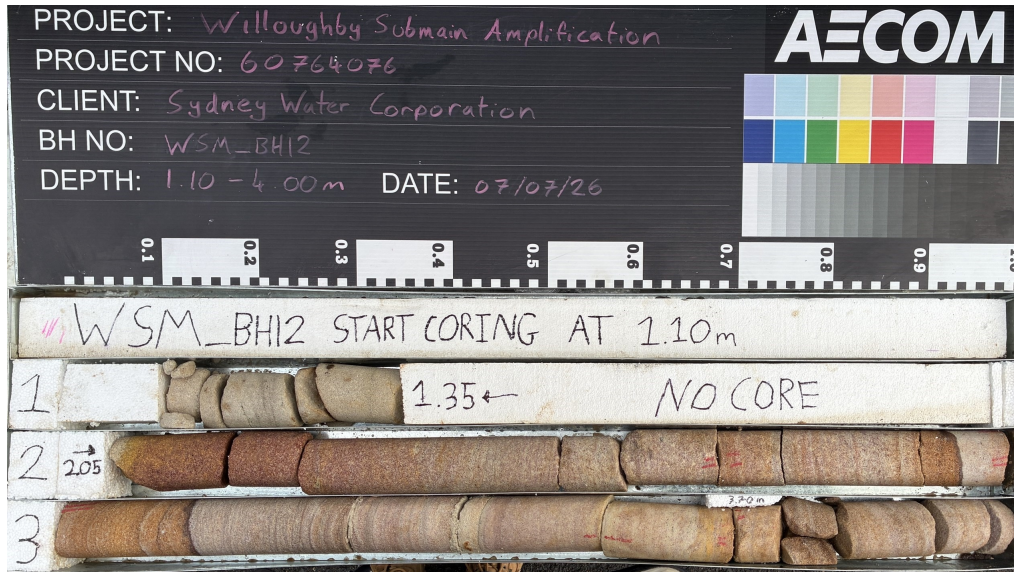
CORE PHOTO REPORT

ID NO. : WSM_BH12

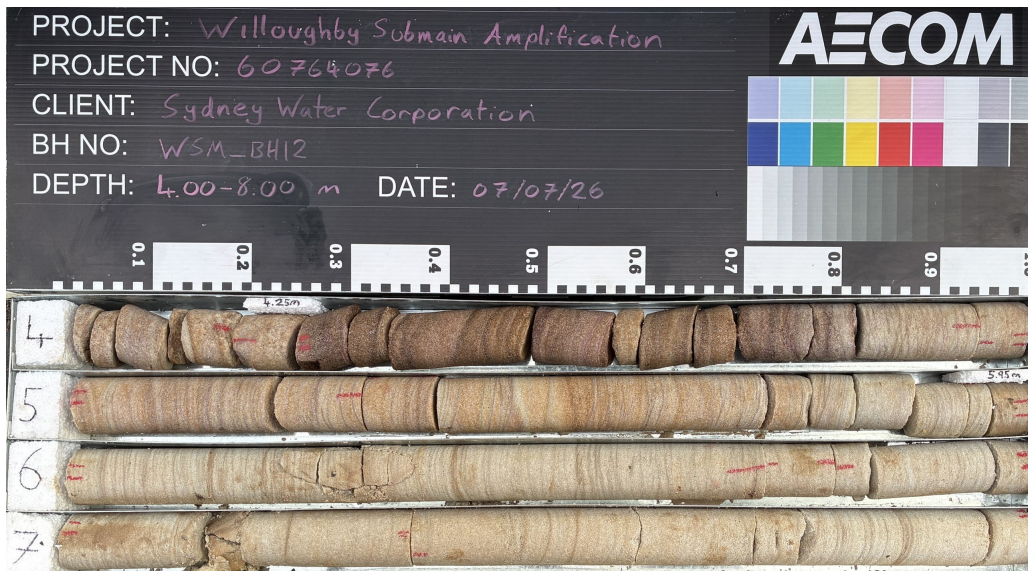
PROJECT NO. : 60764076

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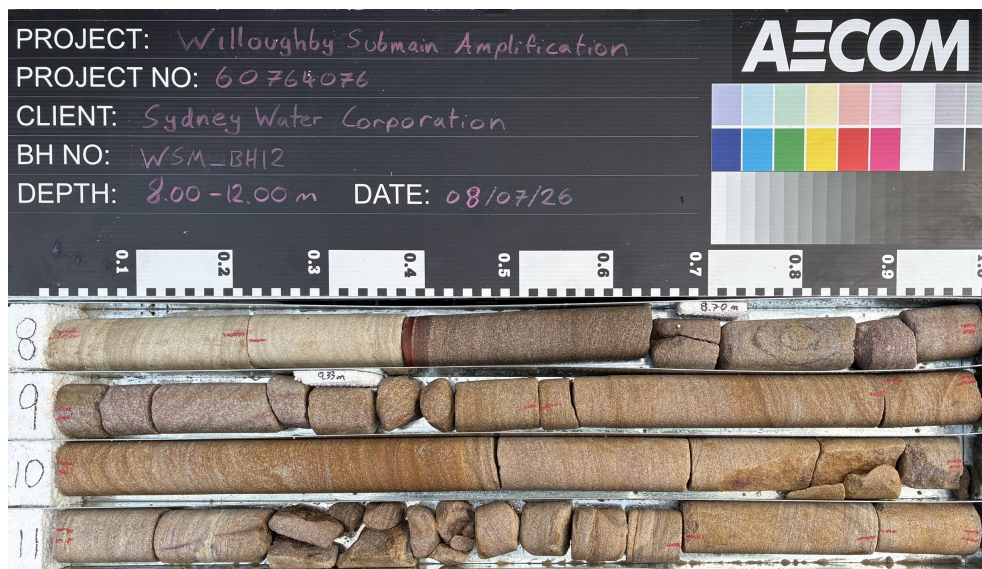
PROJECT : Willoughby Submain Amplification Project
LOCATION : Matthew Street, Crows Nest, NSW 2065



1 - 1.10 to 4.00 m



2 - 4.00 to 8.00 m



3 - 8.00 to 12.00 m

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CORE PHOTO REPORT

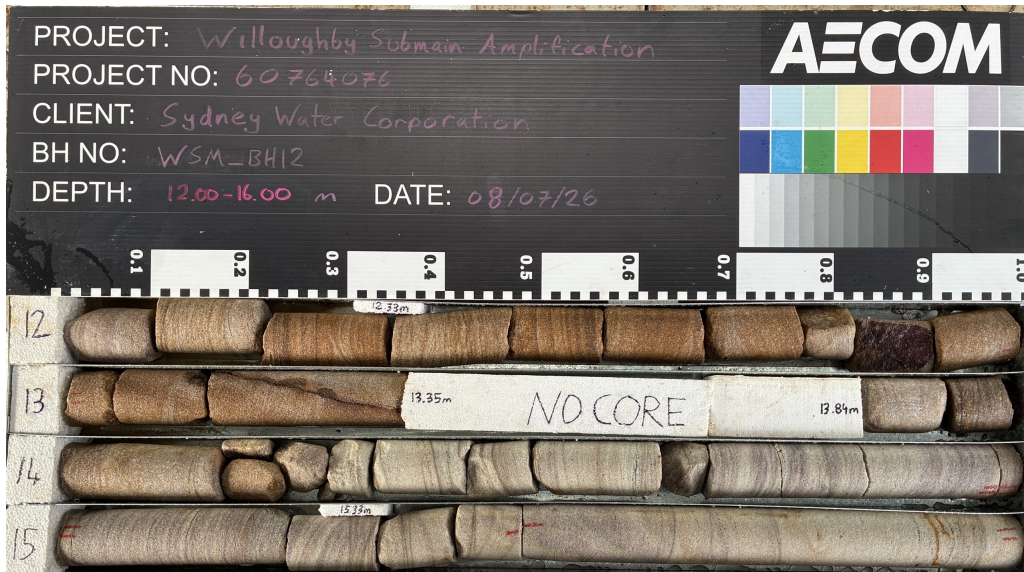
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PROJECT : Willoughby Submain Amplification Project

PROJECT NO. : 60764076

LOCATION : Matthew Street, Crows Nest, NSW 2065

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CORE PHOTO REPORT

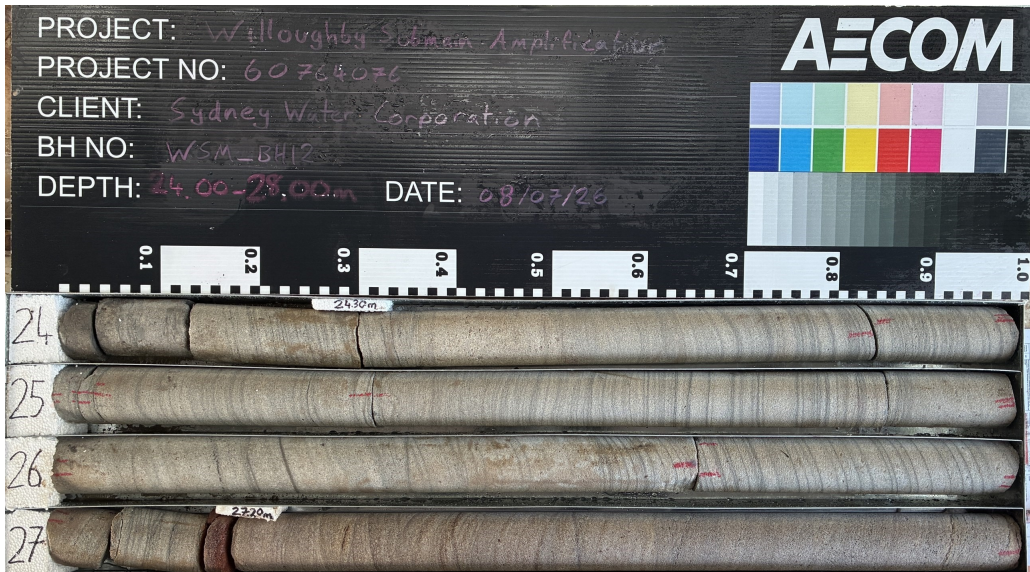
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PROJECT : Willoughby Submain Amplification Project

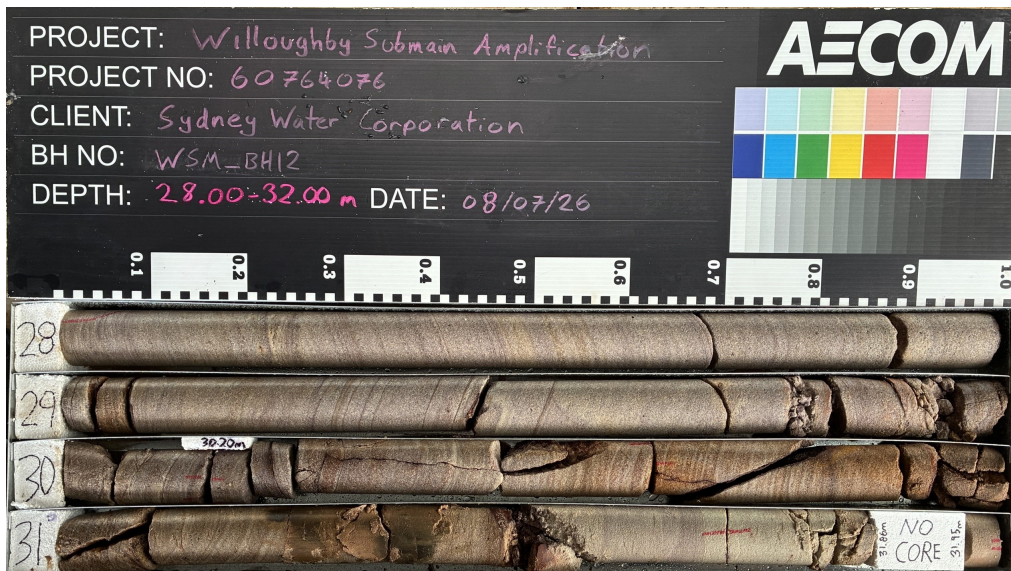
PROJECT NO. : 60764076

LOCATION : Matthew Street, Crows Nest, NSW 2065

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7 - 24.00 to 28.00 m



8 - 28.00 to 32.00 m



9 - 32.00 to 36.00 m

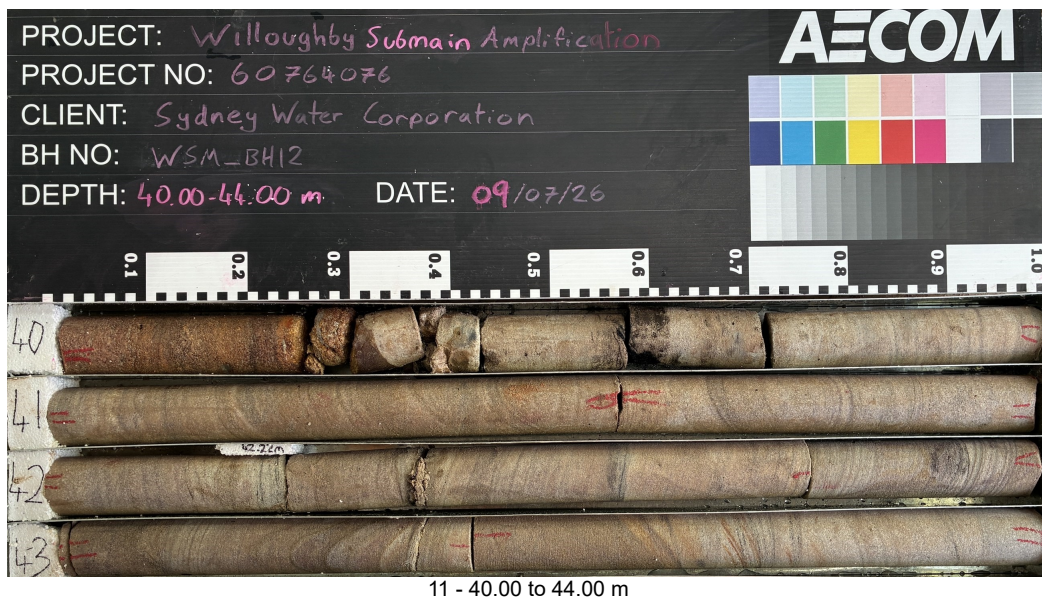
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ID NO. : WSM_BH12

PROJECT : Willoughby Submain Amplification Project
LOCATION : Matthew Street, Crows Nest, NSW 2065

PROJECT NO. : 60764076
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CORE PHOTO REPORT

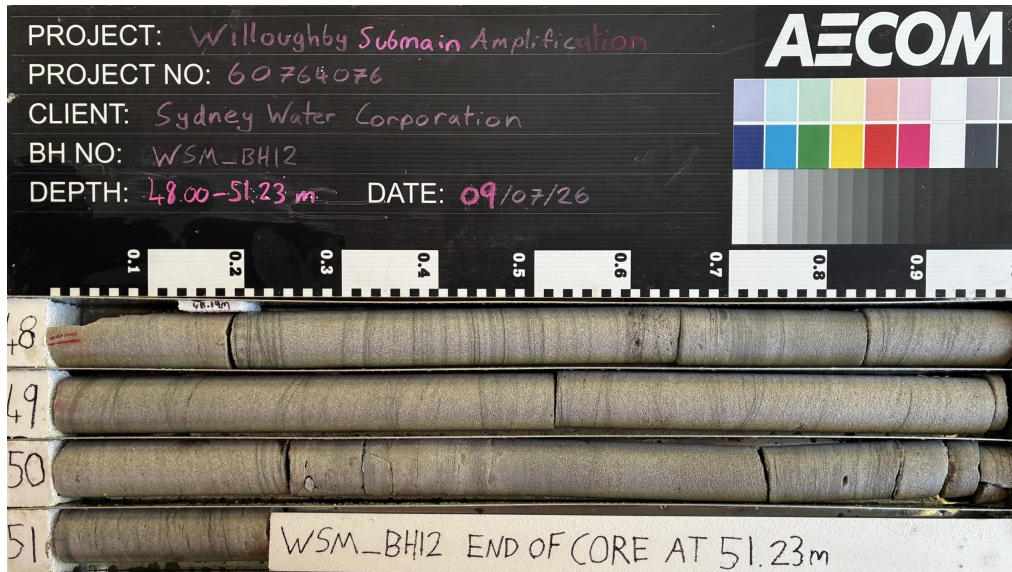
ID NO. : WSM_BH12

PROJECT : Willoughby Submain Amplification Project

PROJECT NO. : 60764076

LOCATION : Matthew Street, Crows Nest, NSW 2065

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